

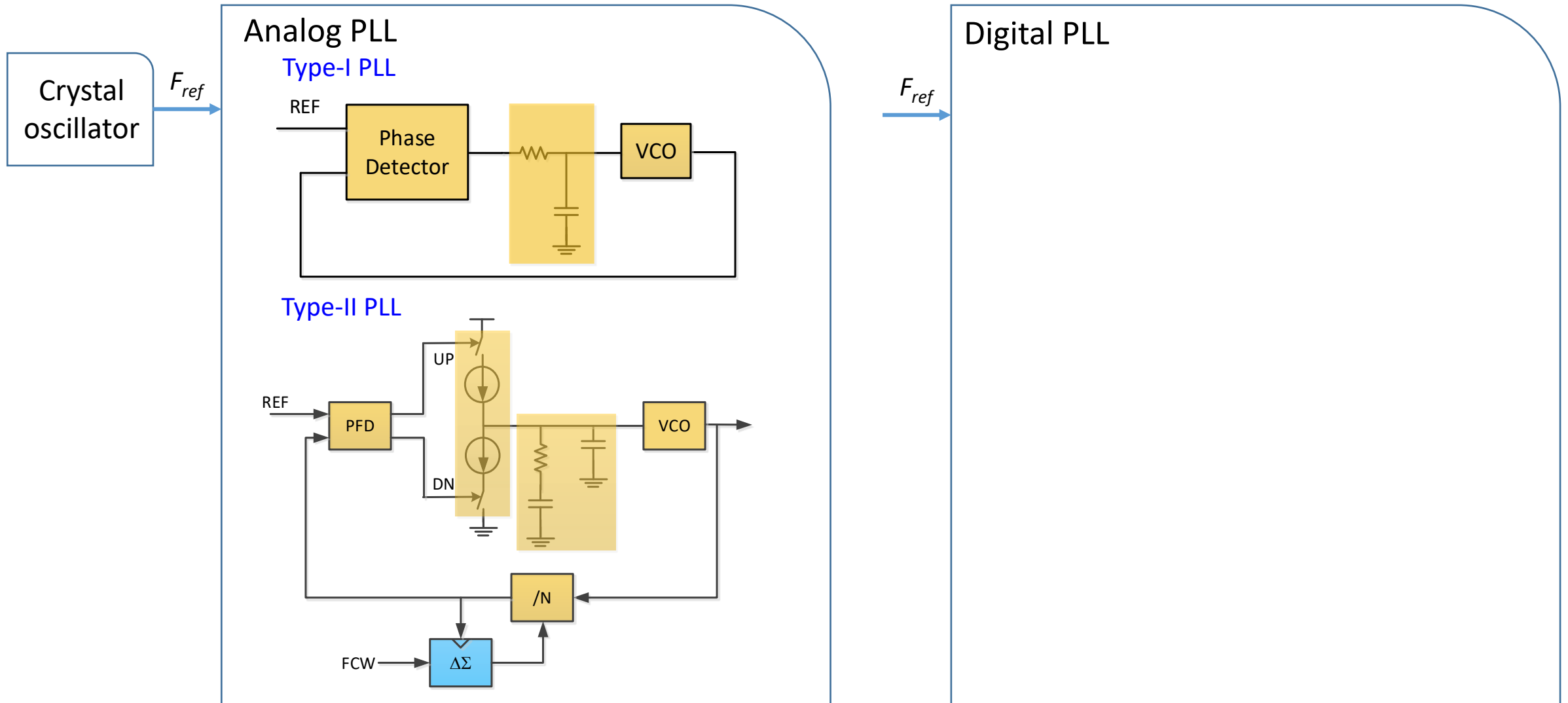
ECE264D: CMOS Analog Integrated Circuits and Systems IV

Lecture 9: Fractional-N Frequency Synthesis

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Frequency Synthesizer Overview



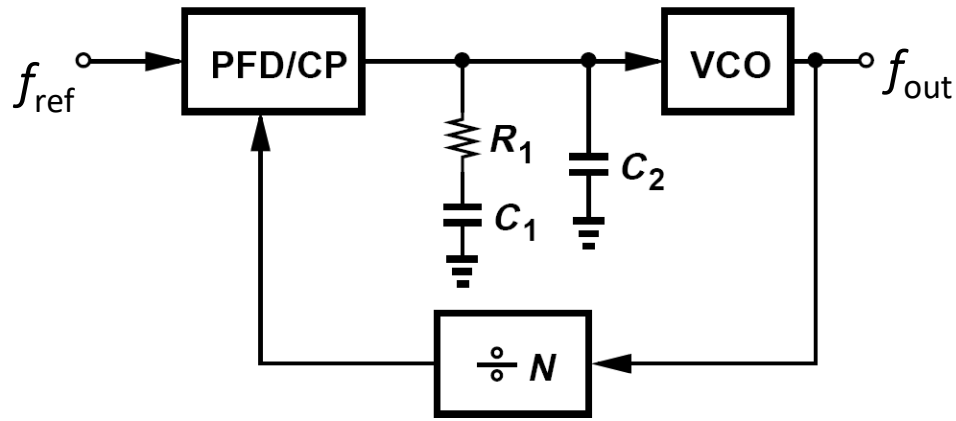
Outline

- Why fractional-N frequency synthesis
- Fractional spurs and randomization
- Quantization noise and noise shaping
- Delta Sigma Modulator (DSM)
- DSM quantization noise in PLL



Concept of Fractional-N Frequency Synthesis

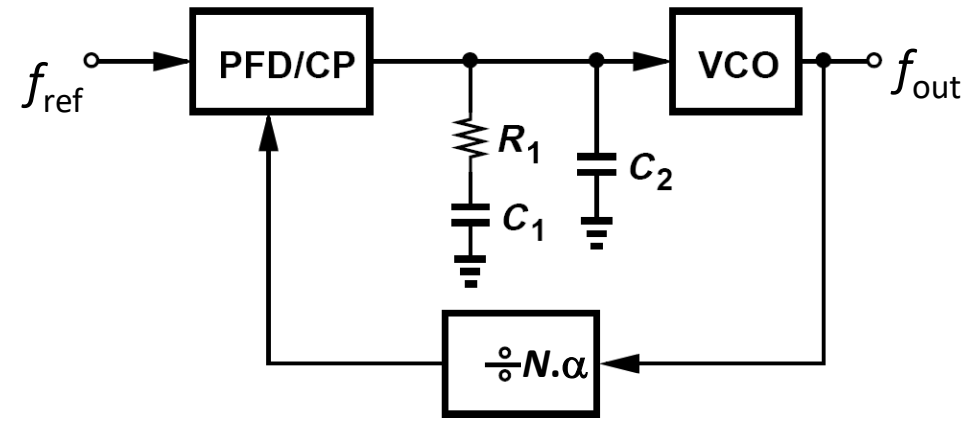
Integer frequency synthesis



After locking, $f_{\text{out}} = N \cdot f_{\text{ref}}$

- Frequency programmability resolution is f_{ref} .
- Reference signal noise is multiplied by $20\log_{10} N$
- Loop bandwidth is limited by $\sim 10\% \cdot f_{\text{ref}}$ for stability

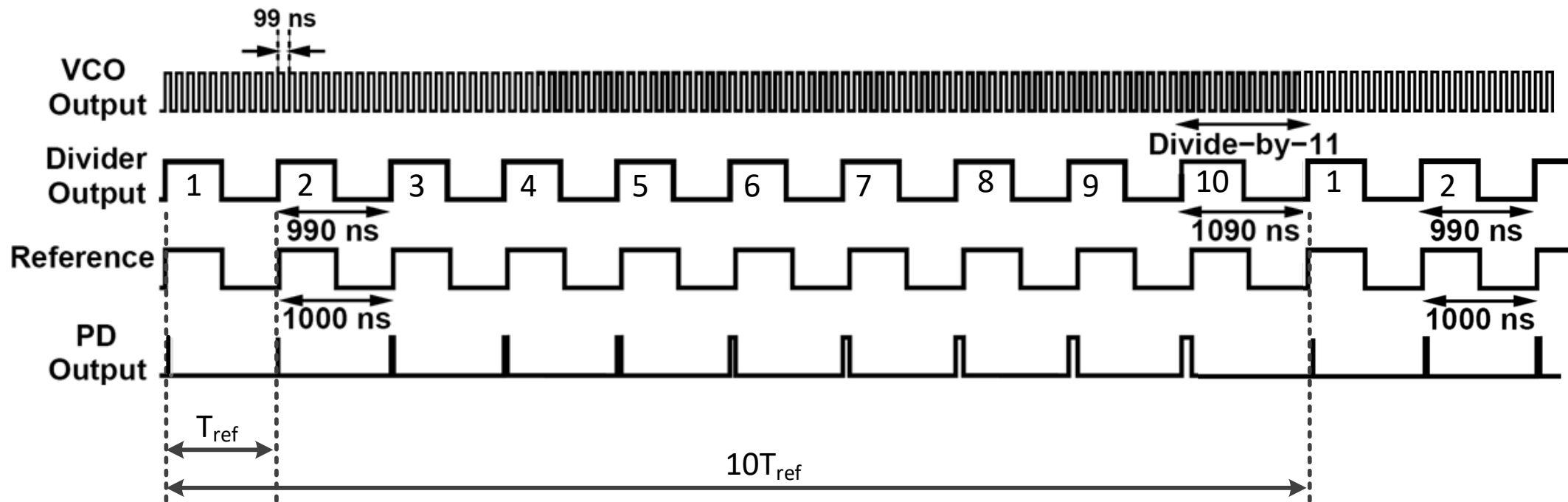
Fractional-N frequency synthesis



After locking, $f_{\text{out}} = N.\alpha \cdot f_{\text{ref}}$

- Frequency programmability resolution is $\ll f_{\text{ref}}$, limited only by the number of digital bits representing α
- Reference signal noise is multiplied by $20\log_{10} N.\alpha$; since f_{ref} can be much larger, $N.\alpha$ is much smaller
- Loop bandwidth can be much wider since f_{ref} can be much larger. Loop bandwidth is limited by quantization noise filtering requirement.

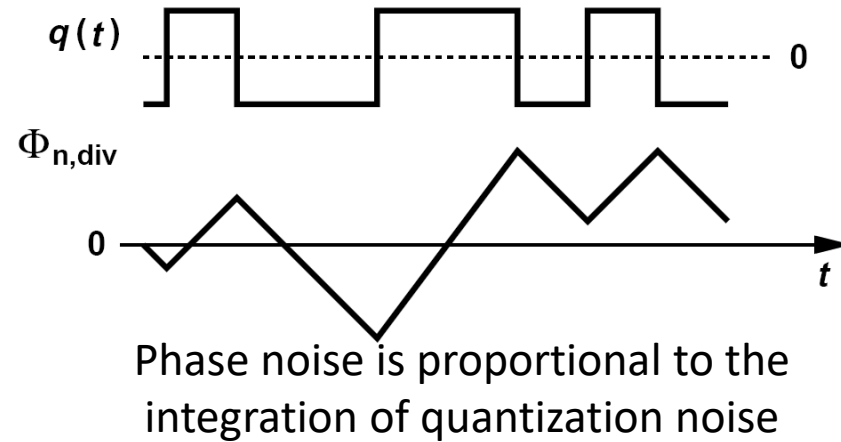
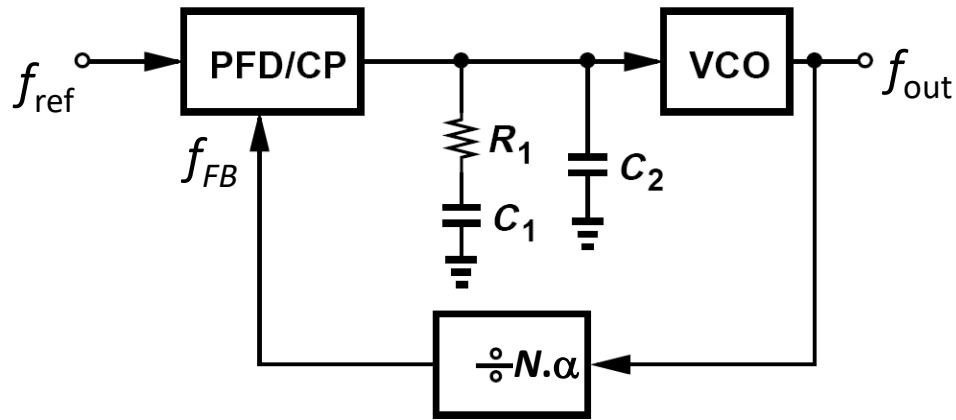
Fractional Divide Ratio Implementation and Fractional Spurs



- Implement fractional divide ratio by averaging
 - $N \cdot \alpha = 10.1$
 - Divide by 10 for 90% of the time (9 reference periods) and divide by 11 for 10% of the time (1 reference period)
- Divider output and PD output have a 'true' period of 10 reference periods $\Rightarrow$ fractional spur at $0.1 \cdot f_{ref}$



Quantization Noise

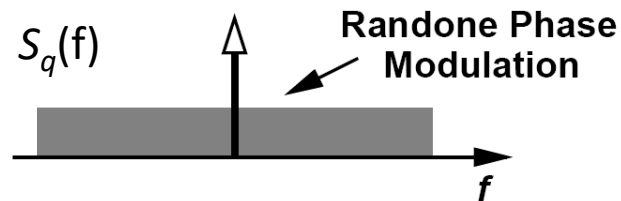


- Let $N+b(t)$ be the instantaneous divide ratio, where $b(t)$ randomly assumes a value of 0 or 1 and has an average of α .
- $b(t) = \alpha + q(t)$, where $q(t)$ is random variable with zero mean: quantization noise
- $$f_{FB}(t) = \frac{f_{out}}{N+b(t)} = \frac{f_{out}}{N+\alpha+q(t)} \approx \frac{f_{out}}{N+\alpha} \left(1 - \frac{q(t)}{N+\alpha} \right) = \frac{f_{out}}{N+\alpha} - \frac{f_{out}}{(N+\alpha)^2} q(t) \quad (1)$$
- Since $f_{ref}(t) = \frac{f_{out}}{N+\alpha}$, PFD output phase error is
$$\phi_n(t) = -\frac{2\pi f_{out}}{(N+\alpha)^2} \int_0^t q(\tau) d\tau \quad (2)$$
- Phase error (or phase noise) is proportional to the integration of quantization noise.



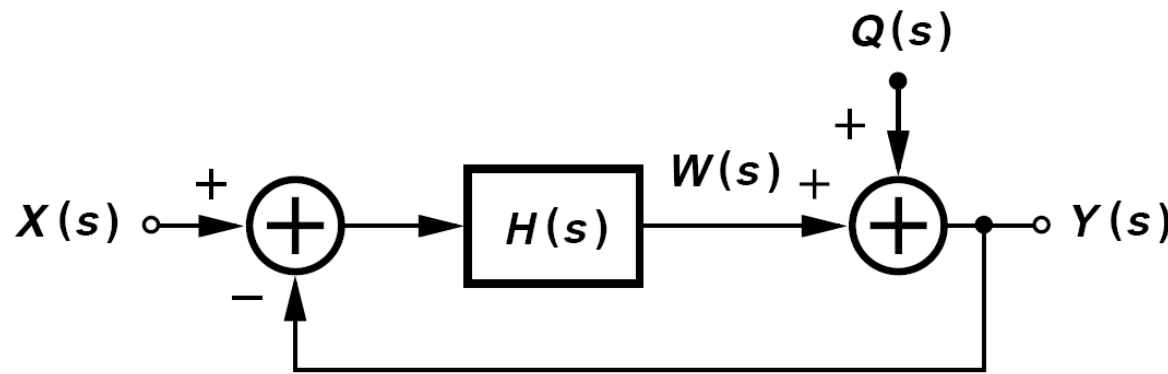
Quantization Noise Induced Phase Noise in Frequency Domain

- $\phi_n(t) = -\frac{2\pi f_{out}}{(N+\alpha)^2} \int_0^t q(\tau) d\tau,$
- $\overline{\Phi_n^2}(f) = \frac{1}{(N+\alpha)^4} \left(\frac{f_{out}}{f}\right)^2 S_q(f),$ where $S_q(f) = \left| \int_{-\infty}^{\infty} q(t) e^{-j2\pi f t} dt \right|^2$ (spectrum of $q(t)$)
- Since $\overline{\phi_{n,out}^2} = (N + \alpha)^2 \overline{\phi_n^2}$ and $f_{out} = (N + \alpha) f_{REF}$
- $\overline{\phi_{n,out}^2} = \left(\frac{f_{REF}}{f}\right)^2 S_q(f)$
- If divide ratio is random between the binary values 0 and 1 with a mean of α , $S_q(f)$ is white noise.



Noise Shaping

- Instead of white noise, what if $S_q(f)$ is high-pass shaped?
 - In-band noise is low
 - Out-of-band noise is high but attenuated by loop filter
 - Yield lower total IPN compared with $S_q(f)$ as white noise
- How to control $b(t)$ or $q(t)$ to obtain high-pass (Note, $b(t) = \alpha + q(t)$)
 - One of the simplest approaches is incorporate a low pass filter in a feedback system
 - The simplest low pass filter is an integrator



- Noise transfer function: High pass

$$\frac{Y(S)}{Q(S)} = \frac{1}{1 + H(s)}$$

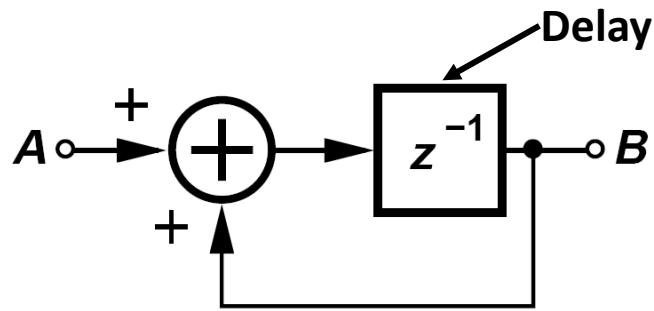
For integrator, $H(s) = \frac{1}{s}$, $\frac{Y(S)}{Q(S)} = \frac{s}{s+1}$

- Signal transfer function: low pass

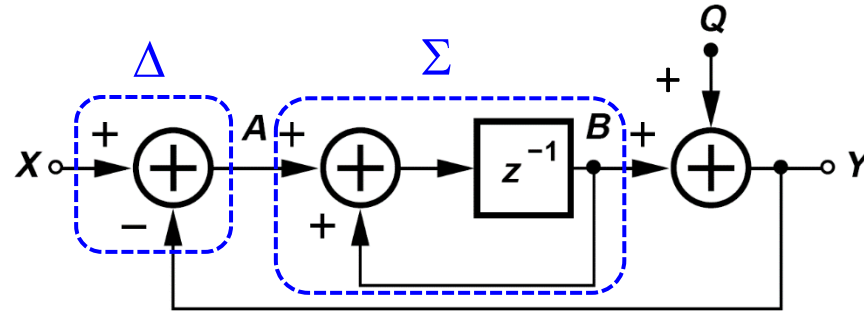
$$\frac{Y(S)}{X(S)} = \frac{H(s)}{1+H(s)} = \frac{1}{s+1} \approx 1 \text{ for low frequency in-band signal}$$



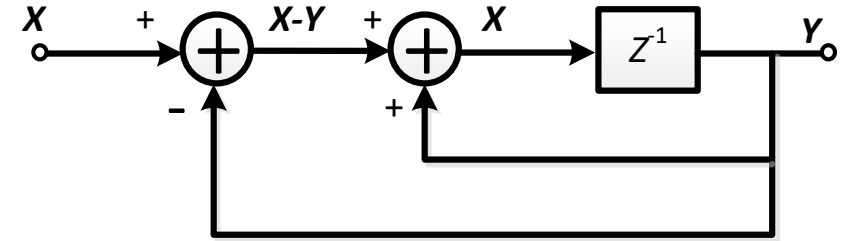
Discrete Time Implementation



Integrator



Integrator in a feedback loop



Signal transfer function

$Q=0$

- Integrator

- $b(n) = a(n) + b(n-1), \frac{B(Z)}{A(Z)} = \frac{z^{-1}}{1-z^{-1}}$

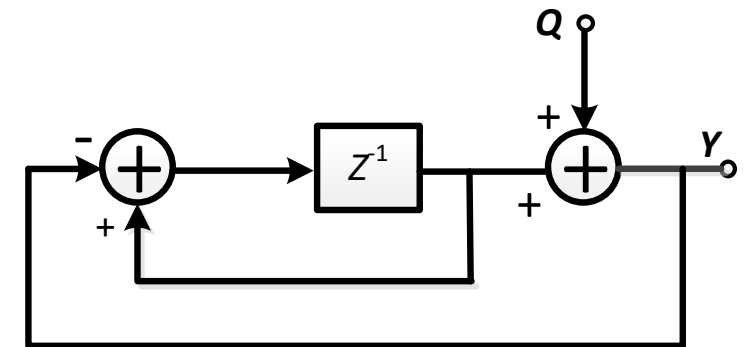
- Integrator in a feedback loop

- Signal transfer function (delay, all pass): $\frac{Y(Z)}{X(Z)} = z^{-1}$

- Quantization noise TF (high pass): $\frac{Y(Z)}{Q(Z)} = 1 - z^{-1}$

- $\Delta\Sigma$ modulator (DSM)

- First order DSM (one integrator)



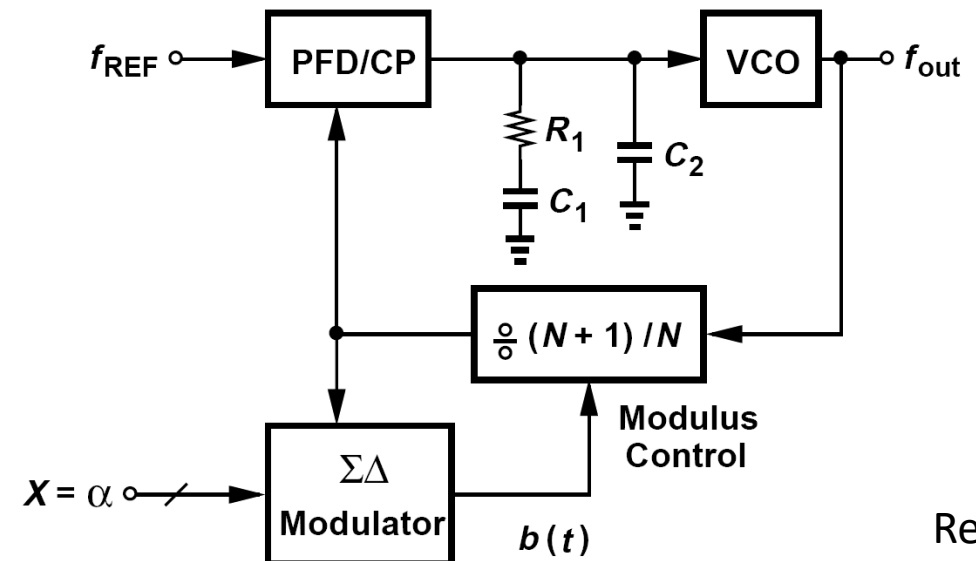
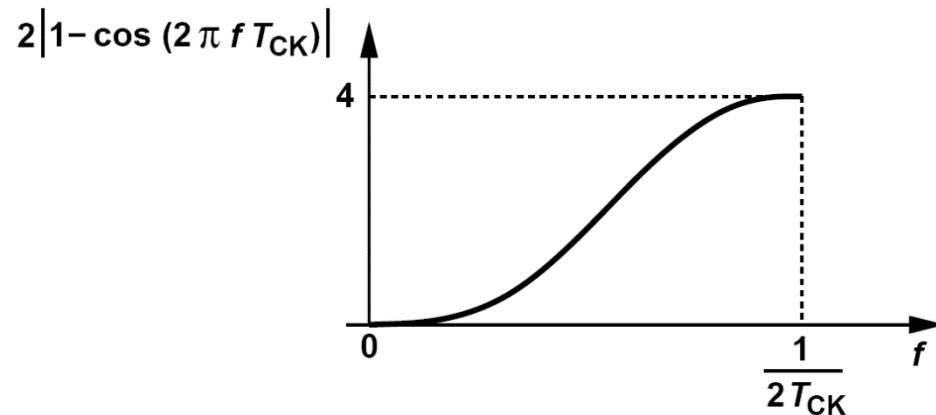
Quantization noise TF

$X=0$



Noise Shaping Function

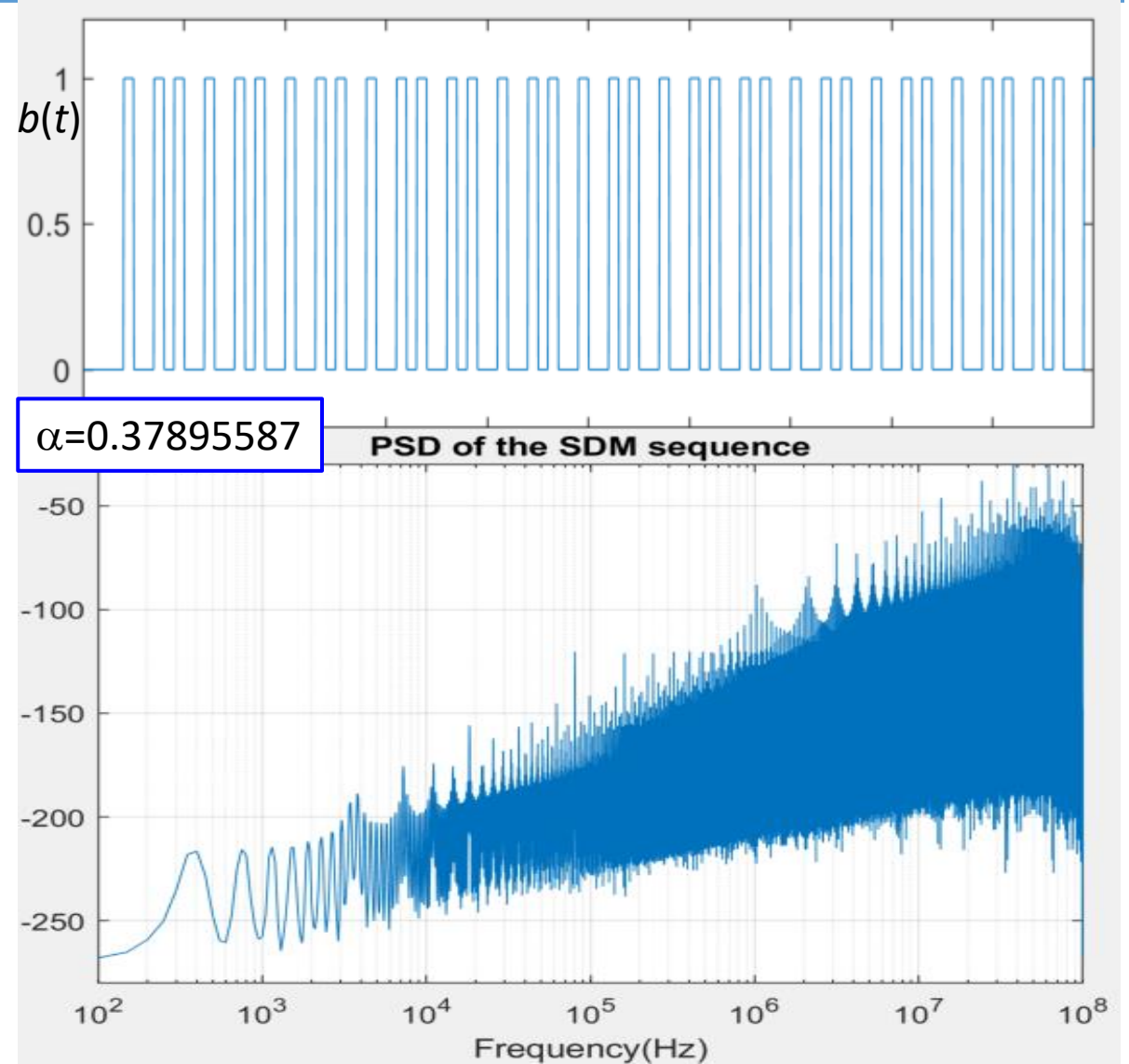
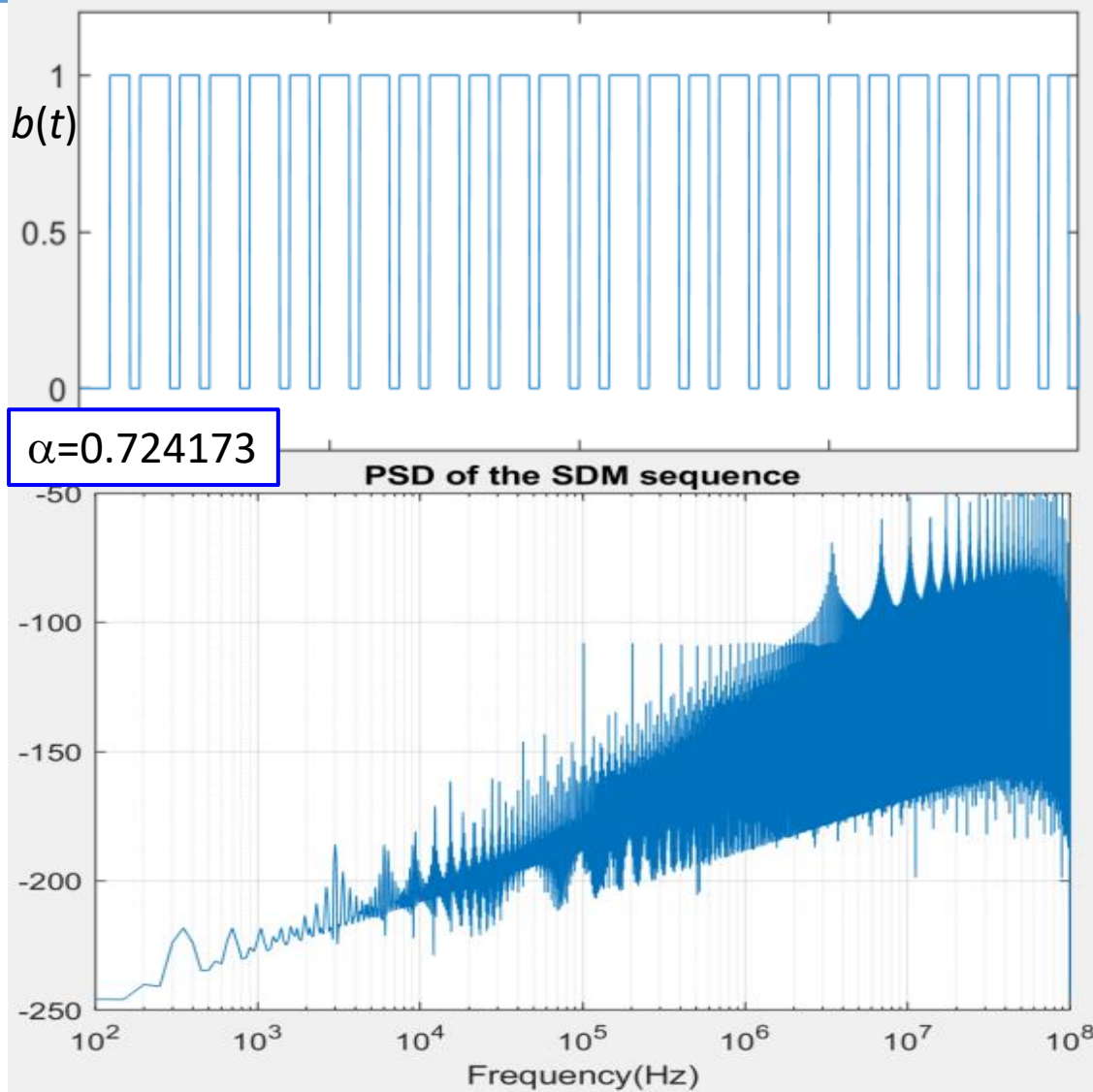
- Quantization noise TF: $\frac{Y(Z)}{Q(Z)} = 1 - z^{-1}$
- $z^{-1} = e^{-sT_{ck}} = e^{-j\omega T_{ck}}$
- $\Rightarrow \frac{Y}{Q}(f) = 1 - e^{-j2\pi f T_{ck}} = e^{-j\pi f T_{ck}}(e^{j\pi f T_{ck}} - e^{-j\pi f T_{ck}}) = 2je^{-j\pi f T_{ck}} \sin(\pi f T_{ck})$
- $\left| \frac{Y}{Q}(f) \right|^2 = |2\sin(\pi f T_{ck})|^2 = 2|1 - \cos(\pi f T_{ck})| \Leftrightarrow$ Noise shaping function
-  The quantization noise $q(n)$ or $Q(f)$ is frequency noise



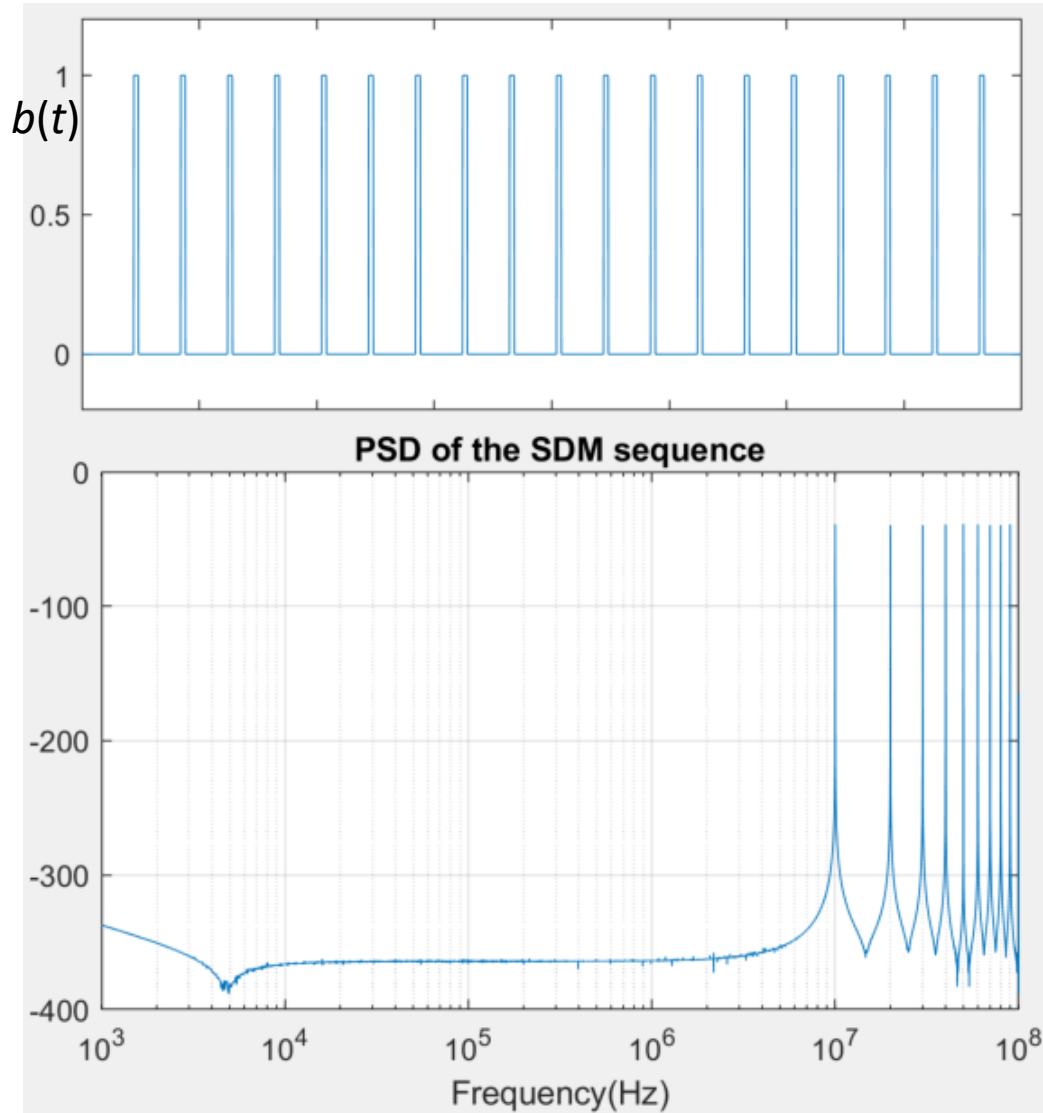
Ref [1]



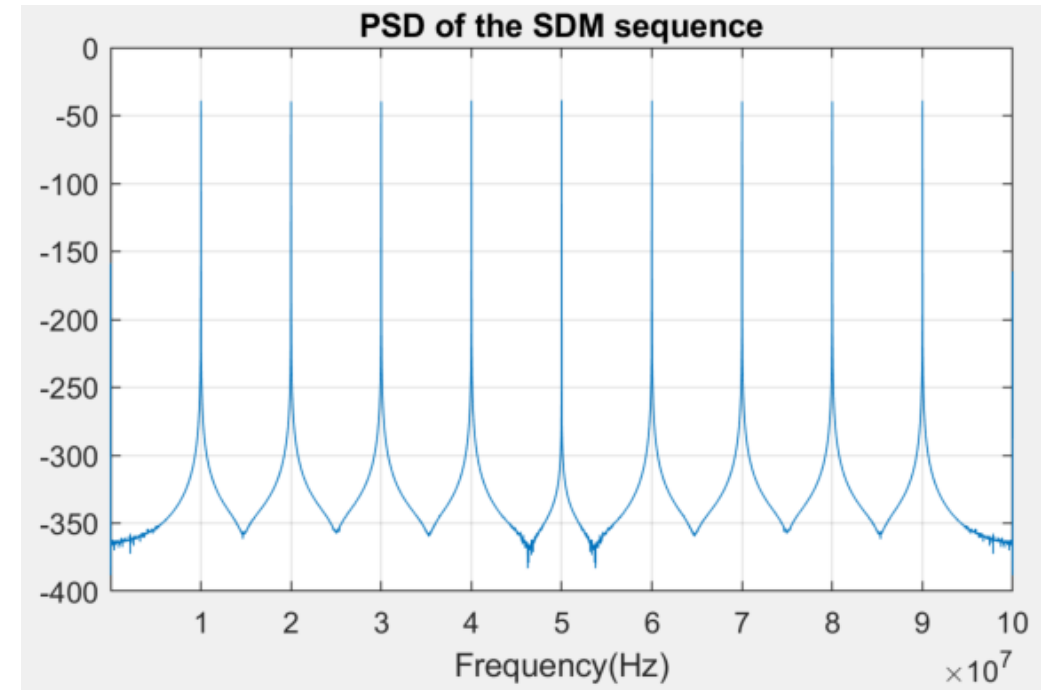
First Order DSM Waveform and Spectrum



First Order DSM Waveform and Spectrum

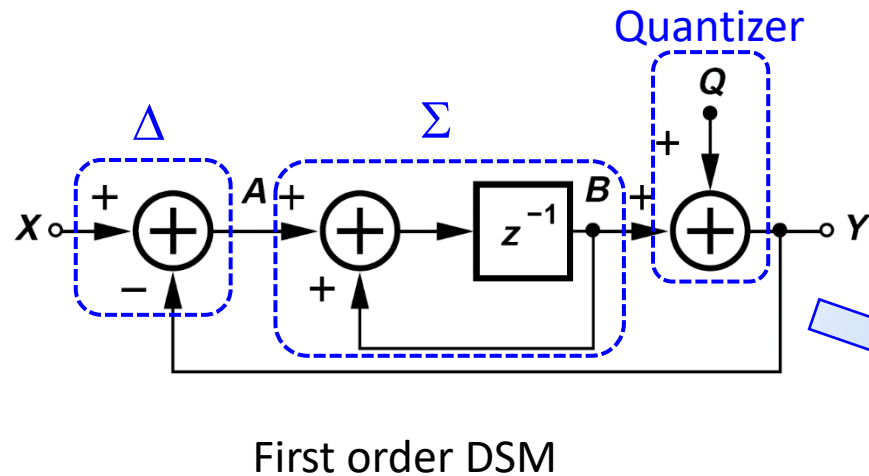


- $\alpha=0.1$
- $N/N+1$ changes periodically
- Strong harmonics at $k \cdot 0.1 \cdot F_{ref}$
- 1st order DSM not sufficient for randomization

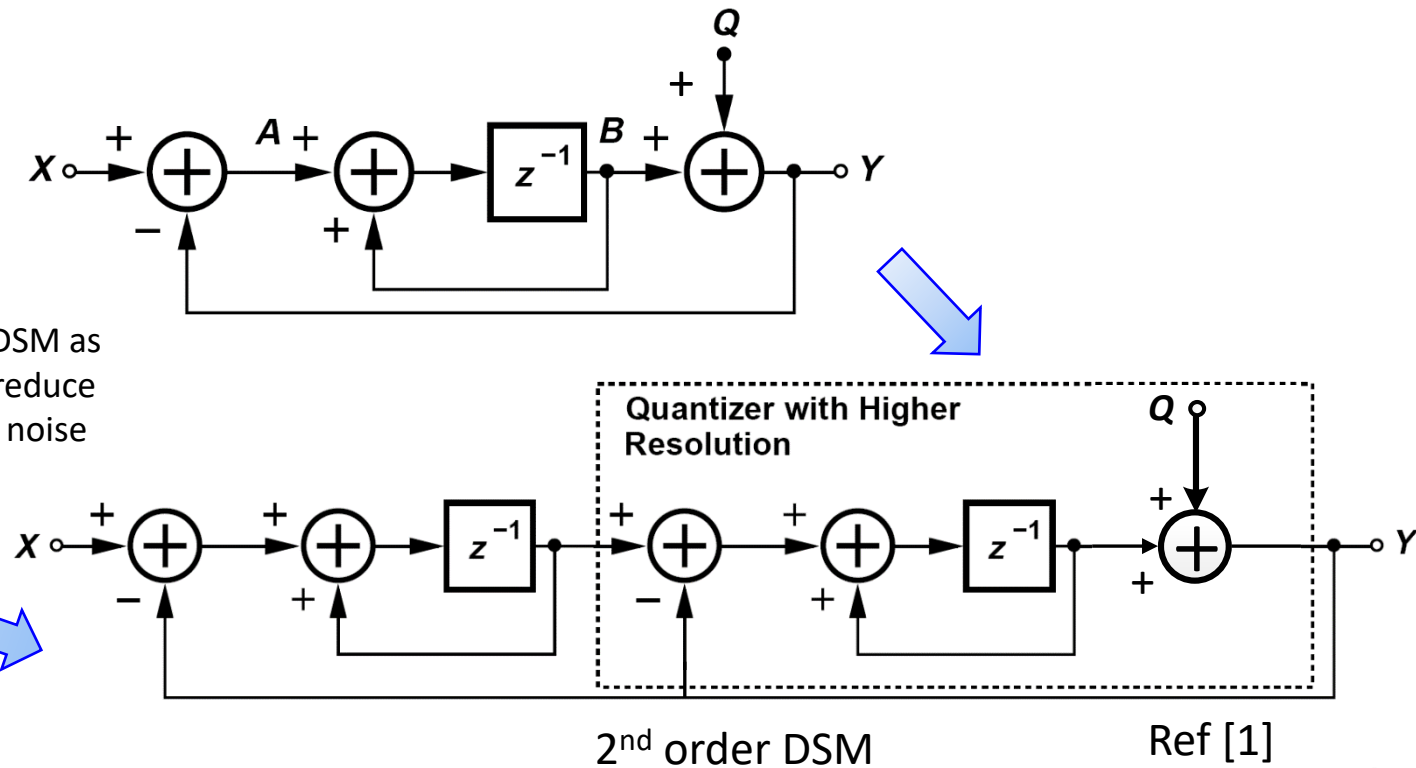


High Order Noise Shaping

- To reduce low frequency in-band quantization noise
 - Increase clock rate that expands the noise shaping function horizontally and reduces noise density at low frequency
 - Increase the noise shaping function slope to push more noise to the high frequency $\Rightarrow$ Higher order DSM
- To improve randomization and reduce sidebands
 - Higher order DSM



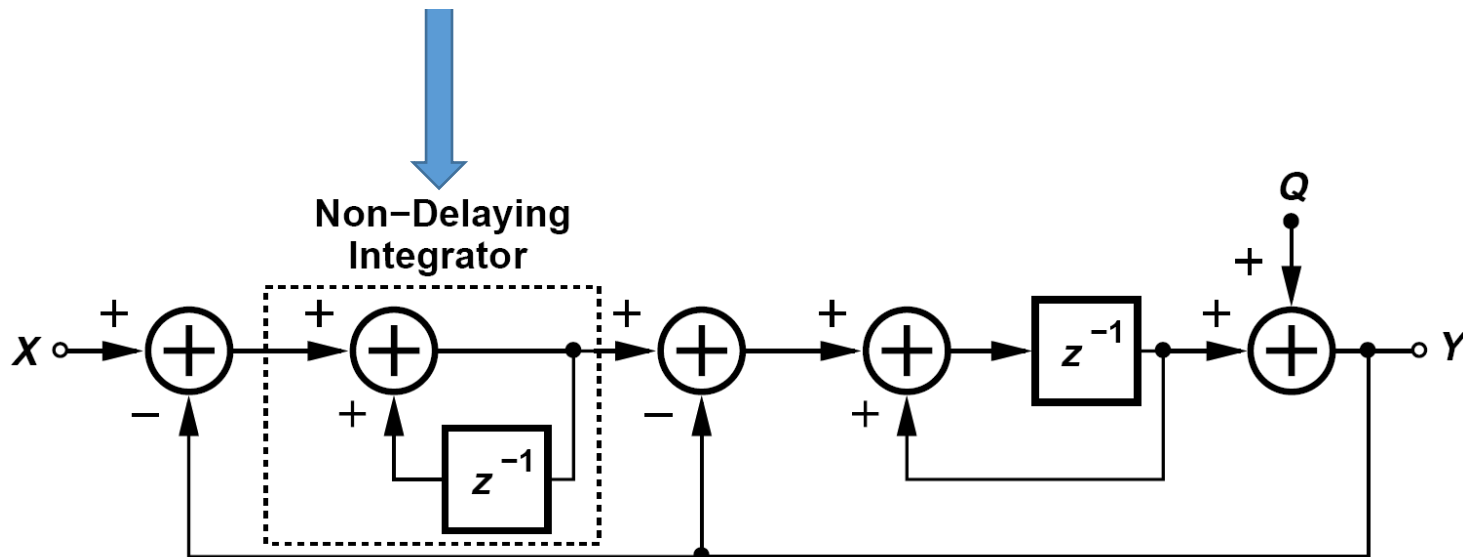
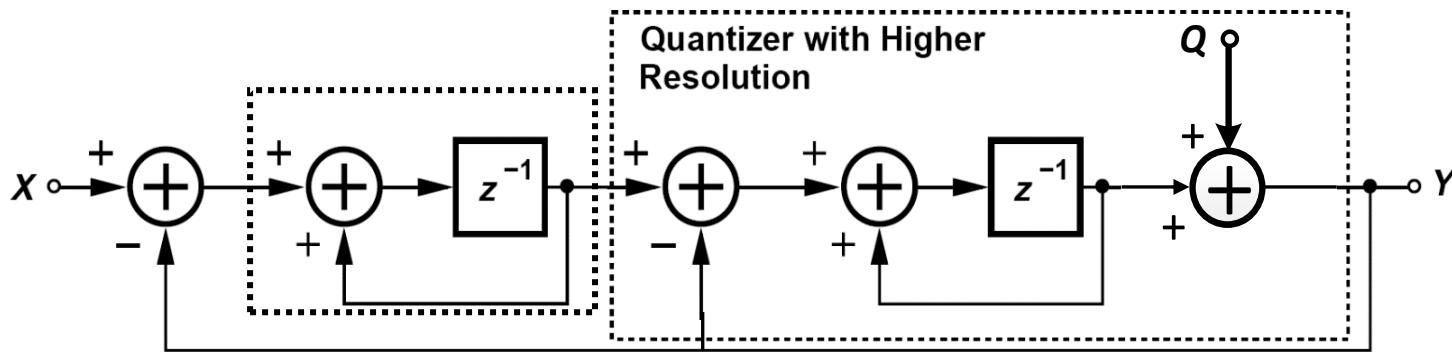
Use another DSM as quantizer to reduce quantization noise



Ref [1]



Second Order DSM Transfer Functions



$$\left(-Y \frac{z^{-1}}{1 - z^{-1}} - Y \right) \frac{z^{-1}}{1 - z^{-1}} + Q = Y$$

$$\frac{Y}{Q}(z) = \frac{(1 - z^{-1})^2}{z^{-2} - z^{-1} + 1}$$

In order not to have poles, replace delaying integrator with non-delaying integrator.

$$\text{Delaying integrator: } H(z) = \frac{1}{z-1} = z^{-1} \frac{1}{1-z^{-1}}$$

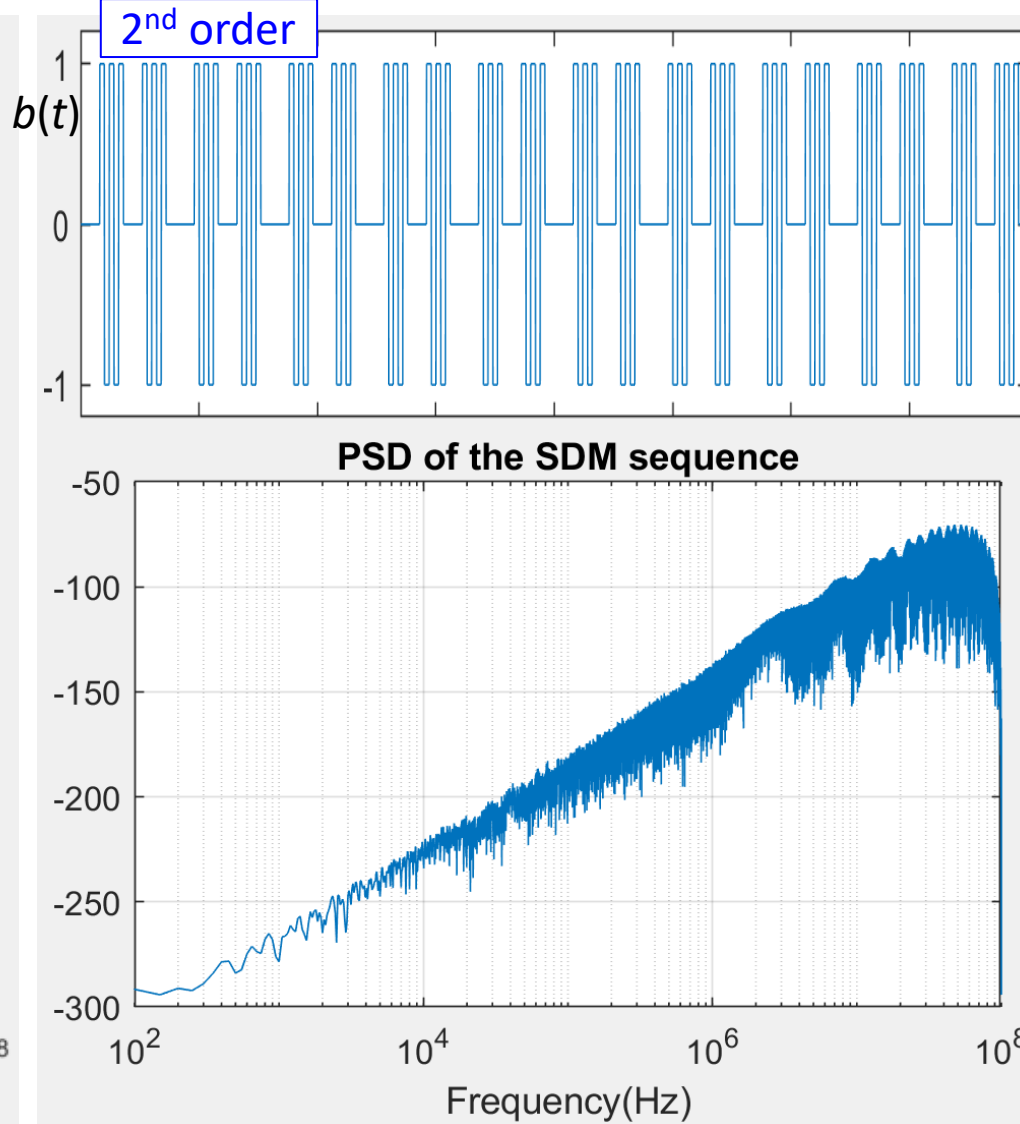
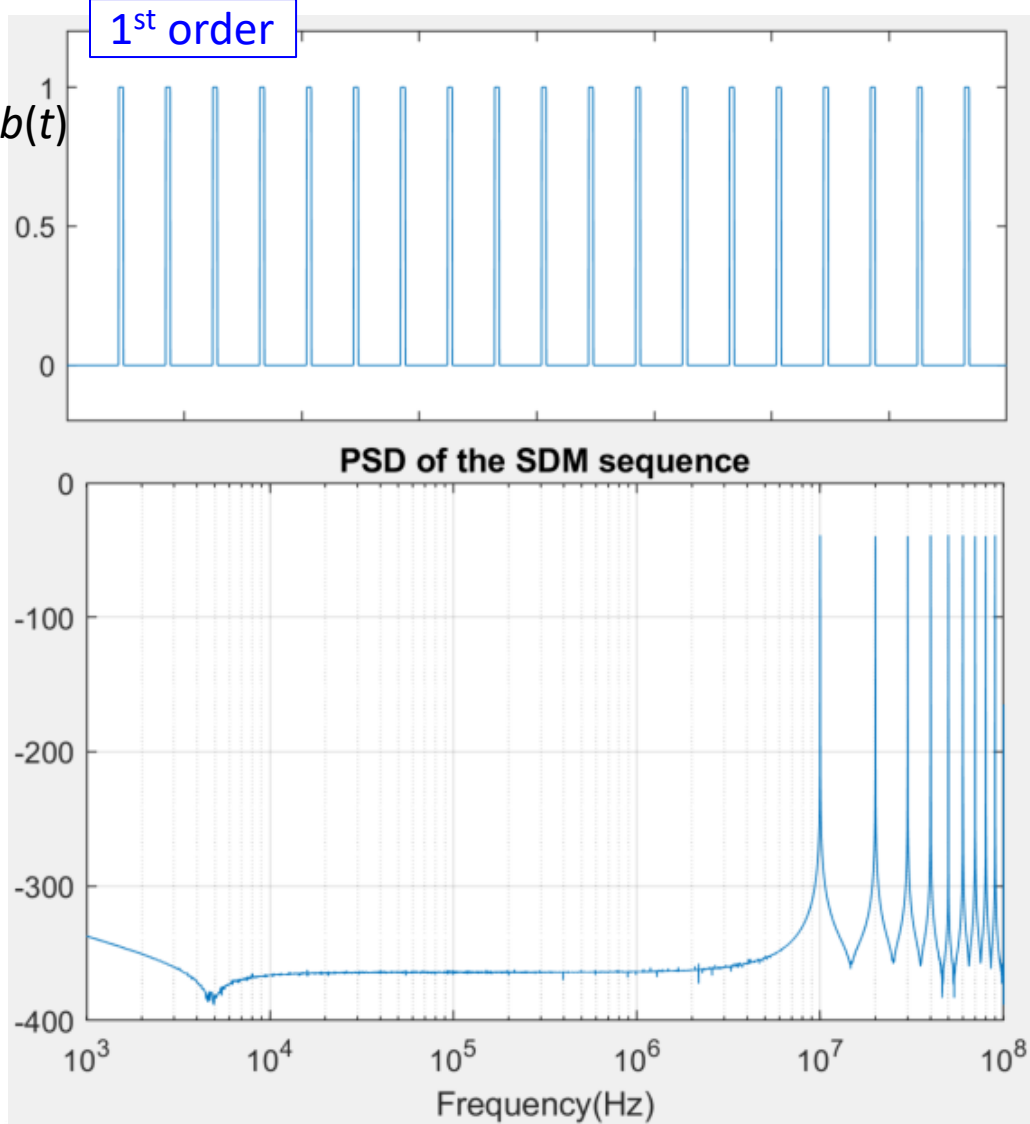
$$\text{Non-delaying integrator: } H(z) = \frac{1}{1-z^{-1}}$$

$$\left(-Y \frac{1}{1 - z^{-1}} - Y \right) \frac{z^{-1}}{1 - z^{-1}} + Q = Y$$

$$\frac{Y}{Q}(z) = (1 - z^{-1})^2$$



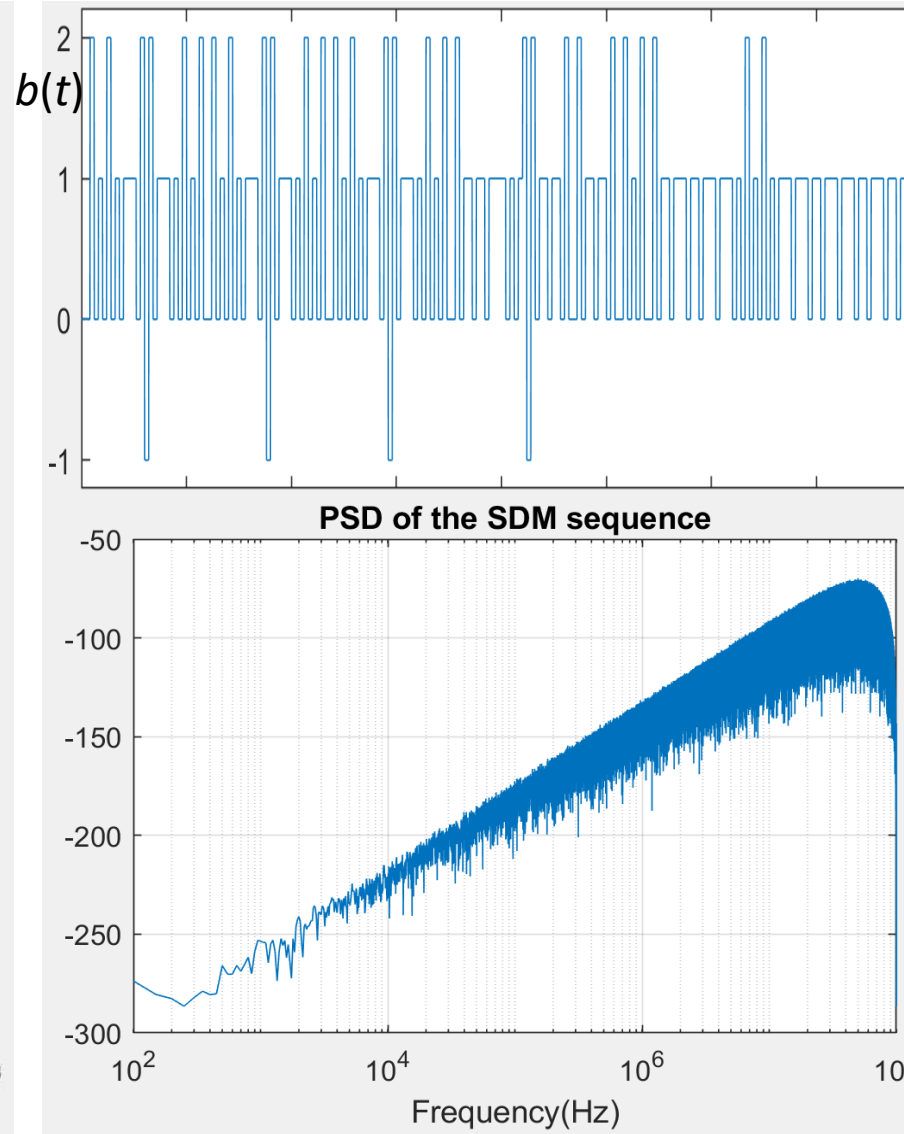
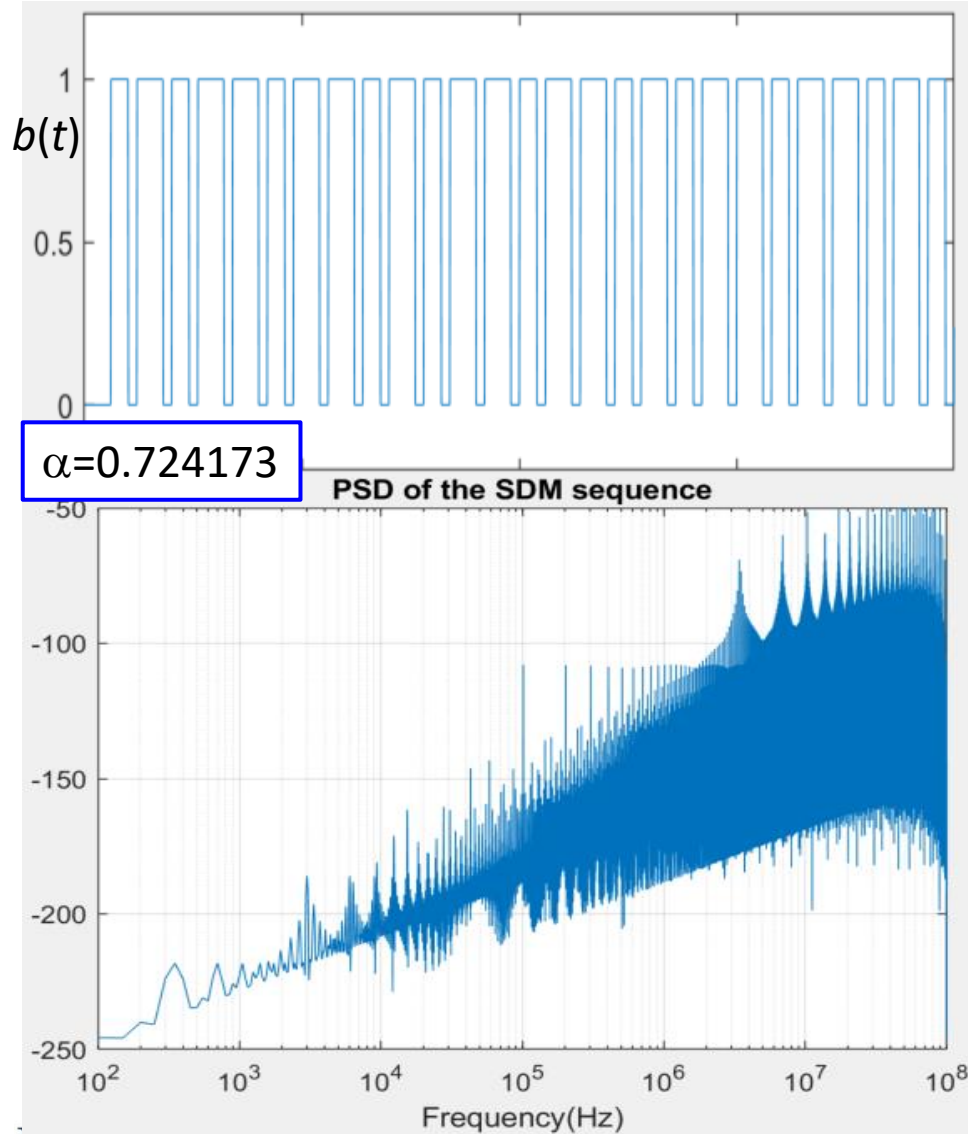
Compare 1st Order and 2nd Order DSM



- $\alpha=0.1$
- 3 levels, still changes periodically
- Less sidebands
- 2nd order DSM more randomization



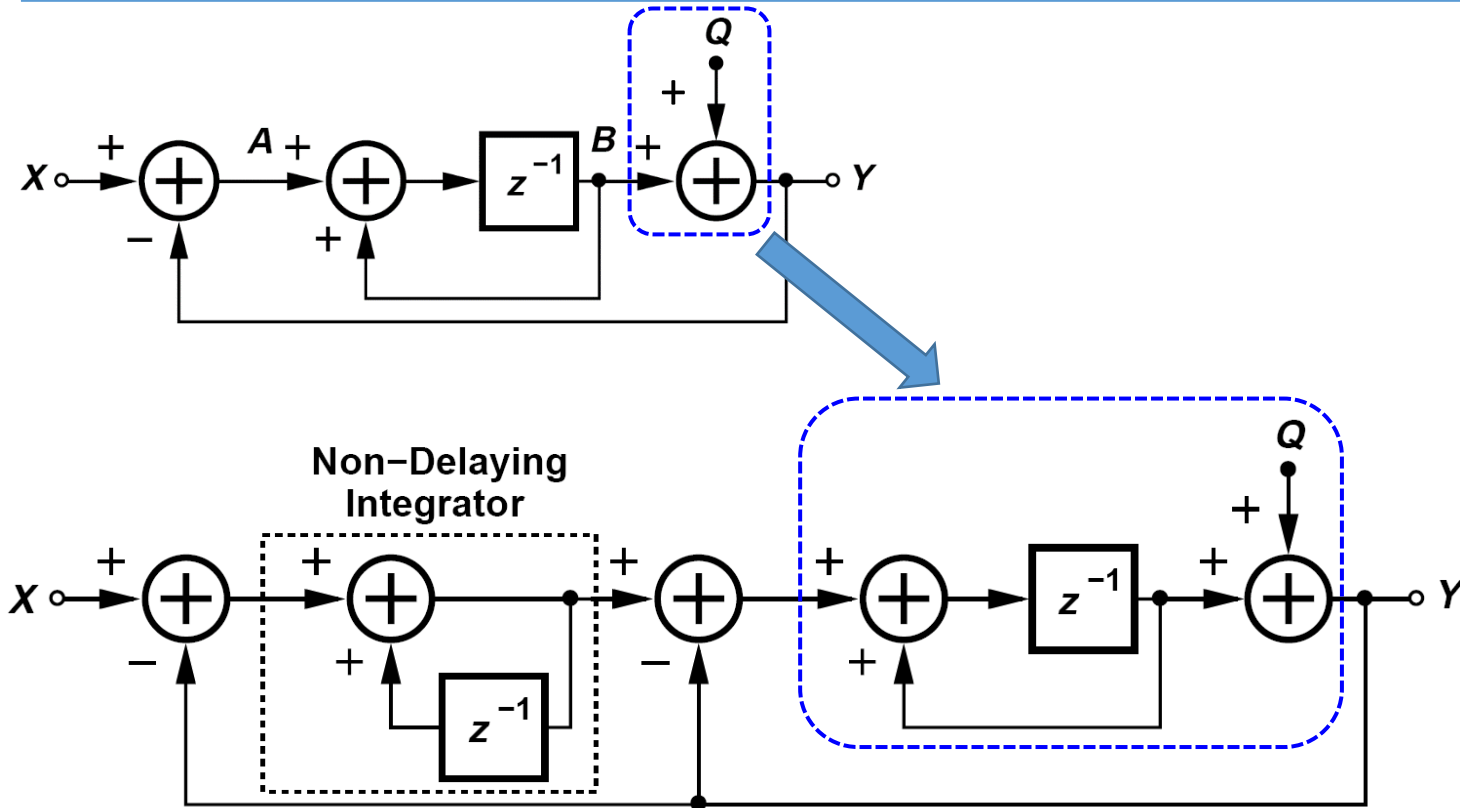
Compare 1st Order and 2nd Order DSM



- $\alpha=0.724173$
- 4 levels
- Less sidebands
- 40dB/dec noise shaping
- Lower in-band noise



Higher Order DSM



- 2nd order DSM is constructed by replacing a simple quantizer with a 1st order DSM that has lower quantization noise.
- Higher order DSM can be constructed similarly by adding another 1st order DSM
 - Single-loop single-quantizer modulators with multi-order filtering in the forward path
- Main issue: stability for 3rd and higher order
- Alternative: Cascade of lower order modulators (multi-stage)
⇒ Multi-Stage Noise Shaping
MASH delta-sigma modulator



MASH Delta Sigma Modulator

- Construct higher order DSM with multi-stage lower order DSM
 - Cascade two or more stable DSM stages
 - Quantization error of each stage is quantized by the succeeding stage and subtracted digitally
 - Order of noise shaping = total orders of each stage
 - No stability issue

$$Y_1(z) = z^{-1}X(z) + (1 - z^{-1})Q_1(z)$$

$$Y_2(z) = z^{-1}Q_1(z) + (1 - z^{-1})Q_2(z)$$

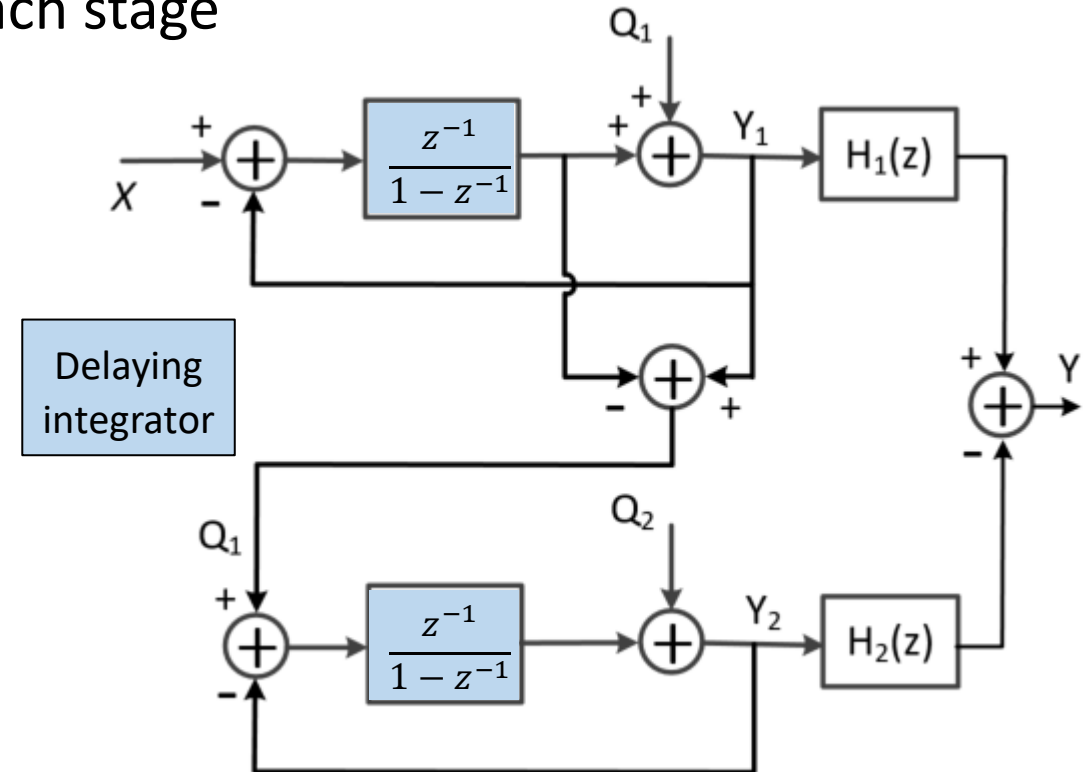
$$Y(z) = H_1(z)Y_1(z) - H_2(z)Y_2(z)$$

If we choose $H_1(z) = z^{-1}$, $H_2(z) = 1 - z^{-1}$

$Q_1(z)$ will be cancelled:

$$Y(z) = z^{-2}X(z) - \underbrace{(1 - z^{-1})^2}_{2^{\text{nd}} \text{ order noise shaping}} Q_2(z)$$

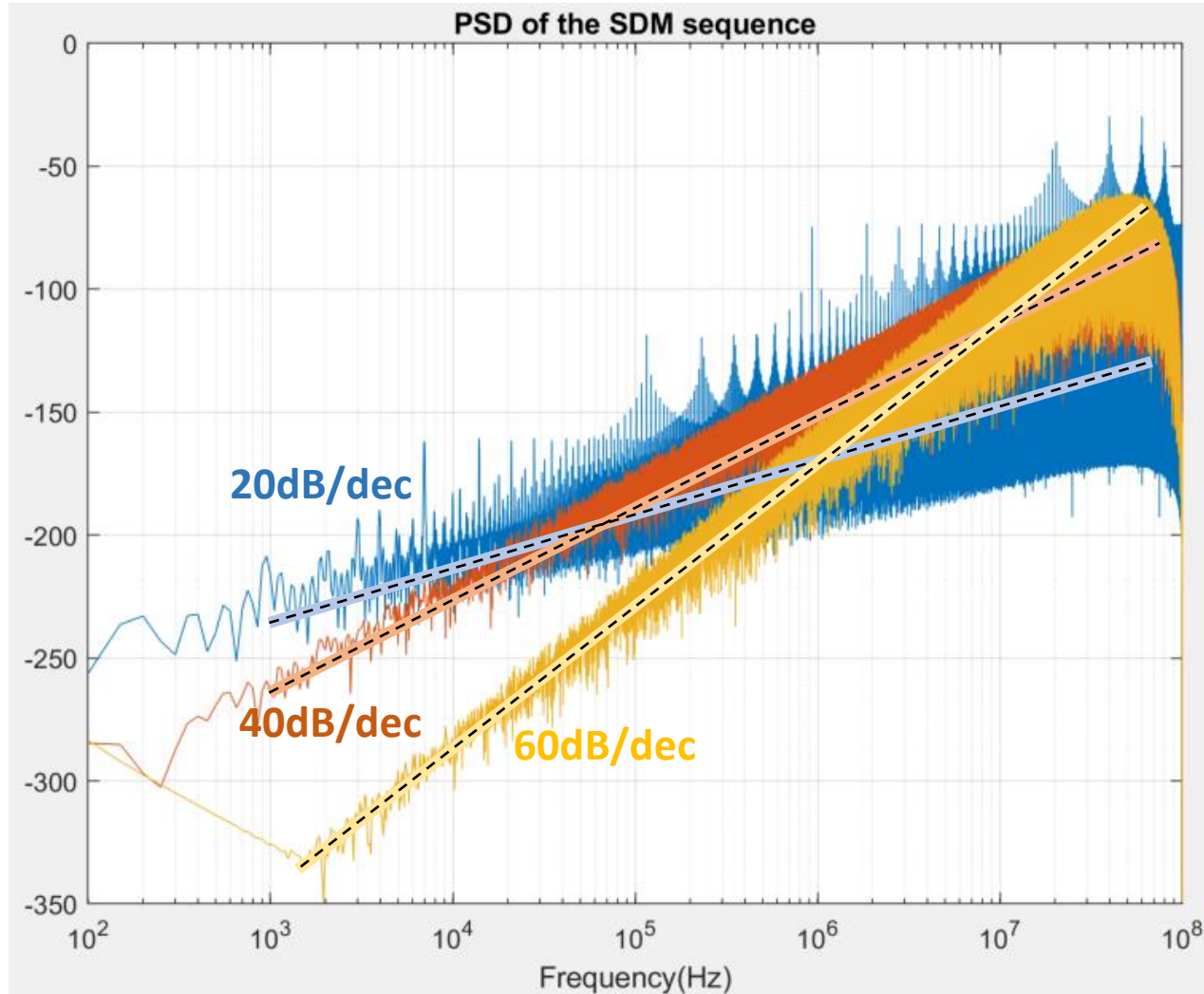
2nd order noise shaping



MASH 1-1



MASH 1, MASH 1-1 and MASH 1-1-1 Noise Shaping Spectrum

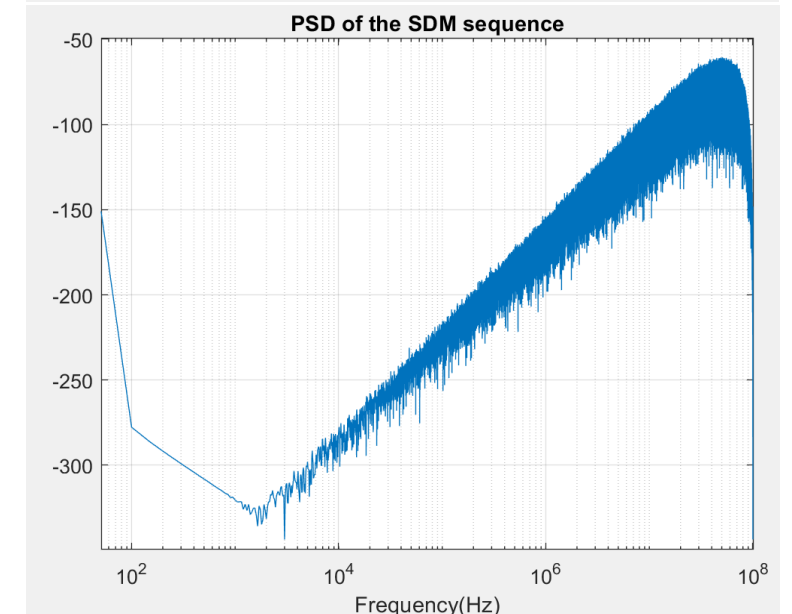


- First order MASH-1 is equivalent to an accumulator
- MASH 1: 20dB/dec
- MASH 1-1: 40dB/dec
- MASH 1-1-1: 60dB/dec
- Lower order, quantization noise is less randomized $\Rightarrow$ More sidebands



Matlab Code for a MSH-111 DSM

```
% Time stamps
Fref=100e6;
Ts=1/Fref;
tindex=2e6;
sim_time=[1:tindex+1].*Ts;
% Loop parameters
alpha=0.724173;
% MASH-111 DSM initialization
acc1 = zeros(1,tindex+2);
acc2 = zeros(1,tindex+2);
acc3 = zeros(1,tindex+2);
s1 = zeros(1,tindex+2);
s2 = zeros(1,tindex+2);
s3 = zeros(1,tindex+2);
c1 = zeros(1,tindex+2);
c2 = zeros(1,tindex+2);
c3 = zeros(1,tindex+2);
p1 = zeros(1,tindex+2);
p2 = zeros(1,tindex+2);
p3 = zeros(1,tindex+2);
y_dsm = zeros(1,tindex+2);
sdm_data_err=y_dsm-alpha;
qn(1)=2;
qn(2)=qn(1)+sdm_data_err(2);
pn(1)=qn(1)*2*pi;
pn(2)=qn(2)*2*pi;
LL=2^23;
d_lsb=1;
if d_lsb==1
    alpha1 = (round(alpha*LL)+round(rand))/LL;
else
    alpha1 = round(alpha*LL)/LL;
end
% DSM MASH-111
for i=1:tindex
    i2 = i+1;
    i3 = i+2;
    % DSM-1
    acc1(i3) = alpha1 + s1(i2);
    s1(i3) = mod(acc1(i3),1);
    c1(i3) = acc1(i3)-s1(i3);
    % DSM-2
    acc2(i3) = s1(i3) + s2(i2);
    s2(i3) = mod(acc2(i3),1);
    c2(i3) = acc2(i3)-s2(i3);
    % DSM-3
    acc3(i3) = s2(i3) + s3(i2);
    s3(i3) = mod(acc3(i3),1);
    c3(i3) = acc3(i3)-s3(i3);
% Combiner
    p1(i3) = c1(i3);
    p2(i3) = c2(i3)-c2(i2);
    p3(i3) = c3(i3)-2*c3(i2)+c3(i);
    % Uncomment following line for MASH-111. Comment it for other modes.
    y_dsm(i3) = p1(i3)+p2(i3)+p3(i3);
    % Uncomment following line for MASH-11. Comment it for other modes.
    %y_dsm(i3) = p1(i3)+p2(i3); % +p3(i3);
    % Uncomment following line for MASH-1. Comment it for other modes.
    %y_dsm(i3) = p1(i3);
    qn(i3)=qn(i2)+y_dsm(i3)-alpha1;% qn: quantization noise
    pn(i3)=qn(i3)*2*pi;% SDM noise in phase domain
end
sdm_data_rec=y_dsm'-mean(y_dsm);
[psd_sdm, fsdm]=pwelch(sdm_data_rec, hann(length(sdm_data_rec)), [],
length(sdm_data_rec), Fref, 'twosided');
figure(1); semilogx(fsdm, 10*log10(psd_sdm));
grid on; xlabel('Frequency(Hz)');
title('PSD of the SDM sequence');
y=y_dsm(1:800);
y1= repmat(y',1,20)';
y1=y1(:)';
figure(2); plot(y1);axis([0, 4000, -3.5, 4.5])
```



References

- 1) Behzad Razavi, RF Microelectronics, Prentice Hall, 2nd edition, 2012
- 2) Richard Schreier and Gabor C. Temes, Understanding Delta Sigma Modulators, Wiley 2005.
- 3) Michael Perrott, M. Trott and C.G. Sodini, A modeling approach for $\Sigma-\Delta$ fractional-N frequency synthesizers allowing straightforward noise analysis, JSSC 2002.

